Introduction to Signals & Communications 2019

Model answers

1. a. i. $f_1(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F_1(\omega) e^{j\omega t} d\omega$

$= \frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi\,\delta(\omega-\omega_0) e^{j\omega t} d\omega$

$\Rightarrow f_1(t) = e^{j\omega_0 t}$ (2)

ii. $f_2(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi\,\delta(\omega+\omega_0) e^{j\omega t} d\omega$

$\Rightarrow f_2(t) = e^{-j\omega_0 t}$ (2)

iii. $\cos(\omega_0 t) = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)$

$\Rightarrow \mathcal{F}[\cos(\omega_0 t)] = \mathcal{F}\left[\frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)\right]$

$\Rightarrow \mathcal{F}[\cos(\omega_0 t)] = \frac{1}{2}\left\{\mathcal{F}[e^{j\omega_0 t}] + \mathcal{F}[e^{-j\omega_0 t}\right.$

$= \frac{1}{2}\left[2\pi\delta(\omega-\omega_0) + 2\pi\delta(\omega+\omega_0)\right]$

$\Rightarrow \mathcal{F}[\cos(\omega_0 t)] = \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)$ (2)

iv. $\cos(\omega_0 t)$ has only frequency of $\pm\omega_0$. (1)

(1)

1. b. i. To show $\cos(m\omega_0 t)$ and $\sin(n\omega_0 t)$ are orthogonal, we consider

$$I_{m,n} = \int_{-\infty}^{\infty} \cos(m\omega_0 t)\sin(n\omega_0 t)\,dt \qquad \forall\ m, n \text{ and } m \neq n$$

Since $\cos(m\omega_0 t)\sin(n\omega_0 t) = \frac{1}{2}\left[\sin((m+n)\omega_0 t) - \sin((m-n)\omega_0 t)\right]$,

$$I_{m,n} = \int_{-\infty}^{\infty} \frac{1}{2}\left[\sin((m+n)\omega_0 t) - \sin((m-n)\omega_0 t)\right]dt$$

As $m-n \neq 0$, the integral above represents integration over two periodic (sine) functions.

Therefore, $I_{m,n} = 0$ for $\forall\ m, n$ & $m \neq n$

That is, $\cos(m\omega_0 t)$ and $\sin(n\omega_0 t)$ are orthogonal to each other. (4)

ii. In fact, all terms of $\cos(n\omega_0 t)$'s and $\sin(m\omega_0 t)$ in the Fourier Series are mutually orthogonal to each other (as part i shows), the energy of $g(t)$ is equal to the sum of the energy of all orthogonal components. That is,

$$P_g = a_0^2 + \sum_{n=1}^{\infty} \frac{a_n^2}{2} + \frac{b_n^2}{2}. \qquad (3)$$

1. b. iii. $g(t)$ can be represented because

- all a_0, $\cos(n\omega_0 t)$, $\sin(n\omega_0 t)$ are mutually orthogonal to each other.
- These components are complete - complete basis for representing $g(t)$ over a period of T.

iv. If $g(t)$ is odd,

$$a_0 = 0 \quad a_n = 0 \quad \forall\, n$$

b_n can be zero or nonzero for some n.

This is so because $\cos(n\omega_0 t)$ for $n = 0, 1, \cdots$ are all even.

Alternatively, when $g(t)$ is odd,

$$g(t) = -g(-t)$$

$$\Rightarrow a_0 + \sum a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)$$

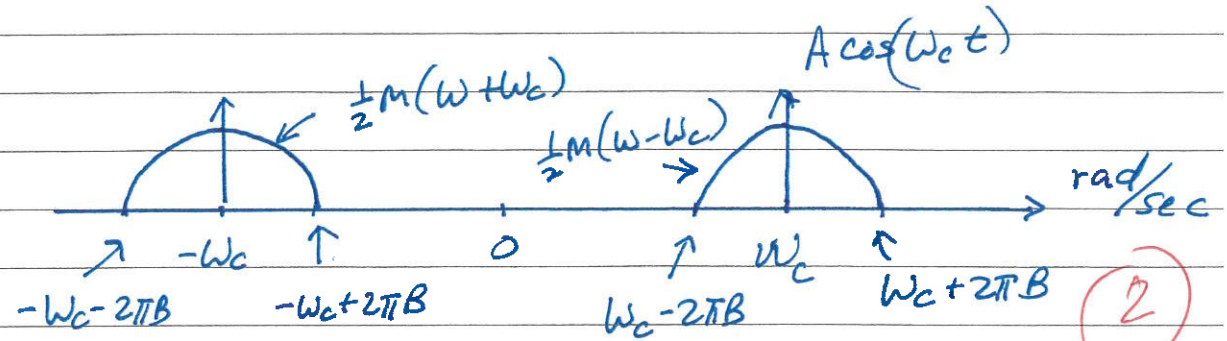
$$= -\left[a_0 + \sum a_n \cos(-n\omega_0 t) + b_n \sin(-n\omega_0 t)\right]$$

$$\Rightarrow 2\left[a_0 + \sum a_n \cos(n\omega_0 t)\right] = 0$$

$$\Rightarrow a_n = 0 \quad \forall \quad n = 0, 1, 2, \cdots$$

1. C. i. $\phi(t) = [A + m(t)]\cos(\omega_c t)$ (2)

ii.



(2)

iii. The bandwidth of $\phi(t)$ is $2B$ Hz. (1)

iv. To use envelop detection, we require

$$A + m(t) \geq 0 \quad \text{for } \forall \text{ time } t.$$

This is so because the envelop of $[A+m(t)]\cos(\omega_c t)$ is identical to $A+m(t)$ when $A+m(t) \geq 0$ for all t. (2)

v. When the carrier signal $A\cos(\omega_c t)$ is removed from $\phi(t)$, the remaining signal is $m(t)\cdot\cos(\omega_c t)$. The energy of the remaining signal using Parseval's theorem is

$$E_{\bar{m}} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{4}\left|M(\omega+\omega_c) + M(\omega-\omega_c)\right|^2 d\omega$$

Since $M(\omega+\omega_c)$ & $M(\omega-\omega_c)$ do not overlap as $\omega_c \gg 2\pi B$,

$$E_{\bar{m}} = \frac{1}{2\pi}\cdot\frac{1}{4}\cdot 2\int_{-\infty}^{\infty}\left[\left|M(\omega+\omega_c)\right|^2 + \left|M(\omega-\omega_c)\right|^2\right]d\omega$$

(3)

$$= \frac{1}{2}\cdot\frac{1}{2\pi}\int_{-\infty}^{\infty}\left|M(\omega)\right|^2 d\omega = \frac{1}{2}\cdot E_m$$

(4)

1. d. i.

$$\varphi(t) = A\cos(\omega_c t + k_f a(t)) \quad (2)$$

ii. $$\hat{\varphi}(t) = A e^{j(\omega_c t + k_f a(t))}$$

$$\Rightarrow \hat{\varphi}(t) = A e^{j k_f a(t)} \cdot e^{j\omega_c t} \quad (3)$$

iii. Use Taylor expansion on $e^{j k_f a(t)}$,

$$\hat{\varphi}(t) = A\left[1 + j k_f a(t) - \frac{k_f^2}{2} a^2(t) + \cdots + j^n \frac{k_f^n}{n!} a^n(t) + \cdots\right] \cdot e^{j\omega_c t}$$

$$\varphi(t) = Re\left[\hat{\varphi}(t)\right]$$

$$= A\left[\cos\omega_c t - k_f a(t)\sin(\omega_c t) - \frac{k_f^2}{2!} \cdot a^2(t)\cos(\omega_c t) + \frac{k_f^3}{3!} a^3(t) \cdot \sin\omega_c t + \cdots\right]$$

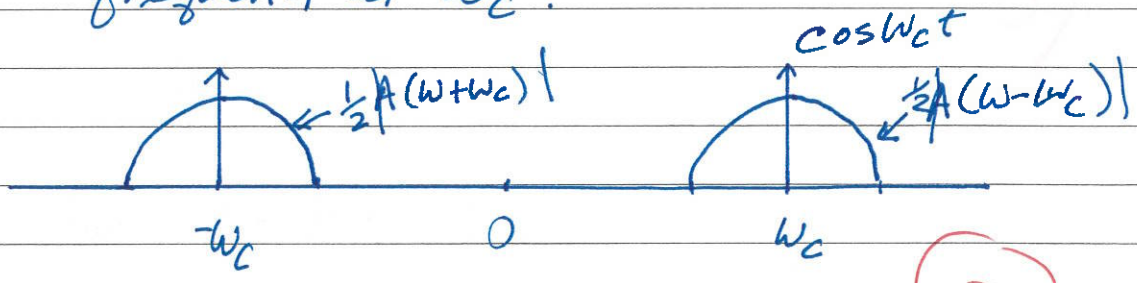
When $|k_f a(t)| \ll 1$, only the first terms in the above are significant. So, (3)

$$\varphi(t) \approx A\left[\cos(\omega_c t) - k_f a(t)\sin\omega_c t\right].$$

1. d. iv. When $|k_f a(t)| << 1$, the carrier angle and thus the frequency change only a small amount. Thus, we have the narrow-band FM signal. (1)

v. As $\varphi(t) \sim A[\cos\omega_c t - k_f a(t) \sin\omega_c t]$,

the first term represents the carrier signal itself. The second term represents a shift of ~~frequency~~ spectrum for $a(t)$ to the frequency at ω_c.



(2)

where $A(\omega) = \mathcal{F}[a(t)]$

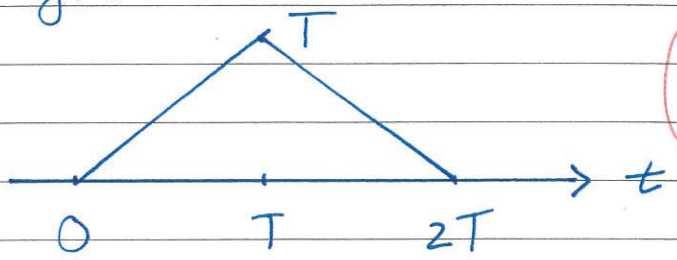
Since $a(t) = \int_0^t m(u)\,du$, $a(t)$ has bandwidth no more than that of $m(t)$, say B Hz. So, the bandwidth of $\varphi(t)$ for $|k_f a(t)| << 1$ is 2B Hz.

2. a. ii.

Case D: $t > 2T$,

$y(t) = 0$ as $x(u)$ & $x(t-u)$ do not overlap. (1)

Graphically, $y(t)$:



(2)

iii. $Y(\omega) = X(\omega) \cdot X(\omega)$

$$Y(\omega) = \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt$$

$$= \int_{t=-\infty}^{\infty} \int_{u=-\infty}^{\infty} x(u)\, x(t-u)\, e^{-j\omega t}\, du\, dt$$

$$= \int_{t=-\infty}^{\infty} \int_{u=-\infty}^{\infty} x(u)\, x(t-u)\, e^{-j\omega(t-u)} \cdot e^{-j\omega u} \cdot du\, dt$$

$$= \int_{u=-\infty}^{\infty} x(u) \int_{t=-\infty}^{\infty} x(t-u)\, e^{-j\omega(t-u)}\, dt \cdot e^{-j\omega u} \cdot du$$

$$\Rightarrow Y(\omega) = \int_{u=-\infty}^{\infty} x(u)\, X(\omega) \cdot e^{-j\omega u}\, du$$

(6)

$$\Rightarrow Y(\omega) = X(\omega) \cdot X(\omega)$$

(8)

2. b. i. $E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$ (2)

ii. $E_g = \int_{t=-\infty}^{\infty} g(t)\, g^*(t)\, dt$

$= \int_{t=-\infty}^{\infty} g(t) \left[\frac{1}{2\pi} \int_{w=-\infty}^{\infty} G(w) e^{jwt} dw \right]^* dt$

$= \frac{1}{2\pi} \int_{t=-\infty}^{\infty} g(t) \int_{w=-\infty}^{\infty} G^*(w) e^{-jwt} dw\, dt$

$= \frac{1}{2\pi} \int_{w=-\infty}^{\infty} \int_{t=-\infty}^{\infty} G^*(w) \cdot g(t) e^{-jwt} dt\, dw$ (4)

$= \frac{1}{2\pi} \int_{w=-\infty}^{\infty} G^*(w) \int_{t=-\infty}^{\infty} g(t) e^{-jwt} dt\, dw$

$\Rightarrow E_g = \frac{1}{2\pi} \int_{w=-\infty}^{\infty} G^*(w)\, G(w)\, dw$

$E_g = \frac{1}{2\pi} \int_{w=-\infty}^{\infty} |G(w)|^2 dw$

2.b. iii $\psi_g(\tau) = \int_{t=-\infty}^{\infty} g(t)\, g(t+\tau)\, dt$ (2)

iv. $\mathcal{F}[\psi_g(\tau)] = \int_{\tau=-\infty}^{\infty} \int_{t=-\infty}^{\infty} g(t)\, g(t+\tau)\, dt\, e^{-j\omega\tau}\, d\tau$

$\Rightarrow \mathcal{F}[\psi_g(\tau)] = \int_{t=-\infty}^{\infty} g(t) \int_{\tau=-\infty}^{\infty} g(t+\tau)\, e^{-j\omega\tau}\, d\tau\, dt$

$= \int_{t=-\infty}^{\infty} g(t) \cdot G(\omega)\, e^{j\omega t}\, dt$

$= G(\omega) \int_{t=-\infty}^{\infty} g(t) e^{j\omega t}\, dt$

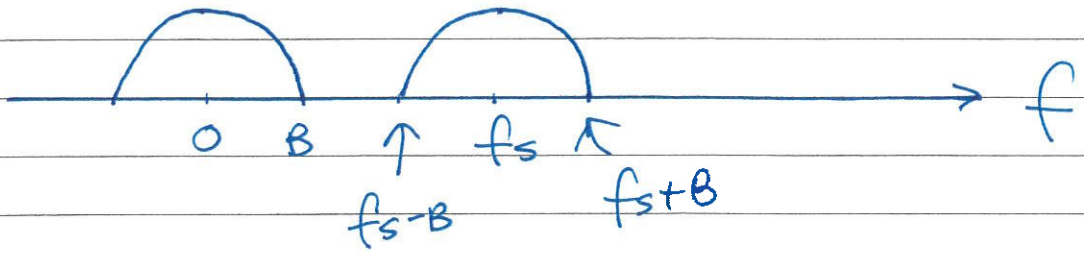
$= G(\omega)\, G^*(\omega)$ $\because g(t)$ is real.

$\Rightarrow \mathcal{F}[\psi_g(\tau)] = |G(\omega)|^2$ (4)

v. $\psi_g(\tau)$ indicates how correlated is $g(t)$ and $g(t+\tau)$. That is, how $g(t)$ and its time shifted version of $g(t)$ by τ are similar to each other. (2)

Usually, $\psi_g(T) = 0$ when $\tau \to \infty$ as $g(t)$ and $g(t+\tau)$ are not related for $\tau \to \infty$. (10)

3. a. i. $f_s \geq 2B$ (4)



The above shows the spectrum of $\tilde{g}(t)$. As a requirement to recover $g(t)$ (at the zero frequency), we require

$f_s - B \geq B$ to avoid replica of spectrum overlapping

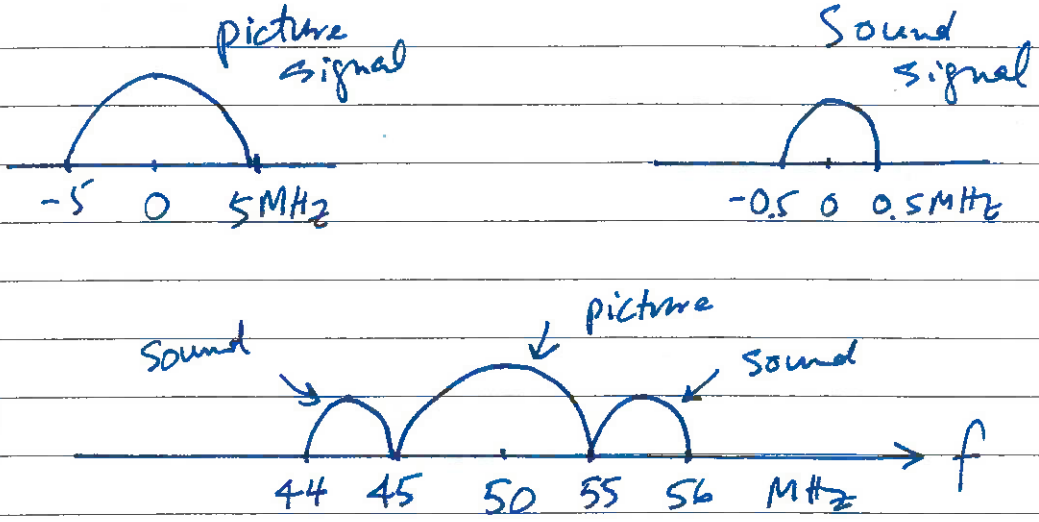
$\Rightarrow f_s \geq 2B$.

ii. The maximum possible bandwidth is $R/2$ Hz because if the bandwidth is less than $R/2$ Hz, we do not need R samples per second to represent the signal. (3)

iii. To transmit a signal of $R/2$ Hz bandwidth, the channel bandwidth $L \geq \frac{R}{2}$. (3)

i.e., $R \leq 2L$, the sample transmission rate must be less than 2 times of channel BW.

3.b. i.



Steps:

a. Modulate the carrier of frequency 5.5 MHz with $m_s(t)$ so that the sound signal is shifted to the frequency band of 5 – 6 MHz

b. Add the picture (baseband) signal to this shifted sound signal.

c. Multiply the combined signal with the carrier of frequency 50 MHz so that the TV signal is transmitted in 44 to 56 MHz frequency band.

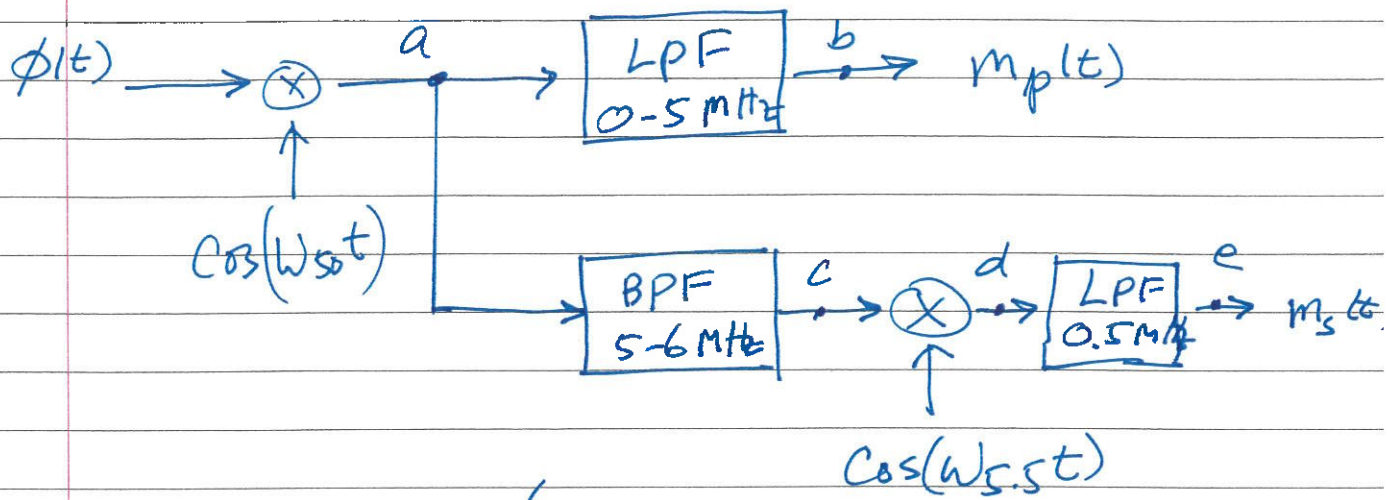
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ii. The TV (transmitted) signal is

$$\phi(t) = \left[m_p(t) + m_s(t) \cdot \cos\left(2\pi \cdot 5.5 \times 10^6 t\right) \right] \cdot \cos\left(2\pi \cdot 50 \times 10^6 t\right)$$

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3. b. iii



$\omega_{50} = 2\pi \times 50 \times 10^6$

$\omega_{5.5} = 2\pi \cdot 5.5 \times 10^6$

$\omega_c = 2\pi \cdot 6 \times 10^6$

(b)

$$\phi(t) = \left[m_p(t) + m_s(t) \cdot \cos(\omega_{5.5} t) \right] \cos(\omega_{50} t)$$

The signal at point a,

$$S_a(t) = \phi(t) \cdot \cos(\omega_{50} t)$$

$$= \left[m_p(t) + m_s(t) \cdot \cos(\omega_{5.5} t) \right] \cos^2(\omega_{50} t)$$

$$= \left[m_p(t) + m_s(t) \cos(\omega_{5.5} t) \right] \cdot \frac{1 + \cos(\omega_{100} t)}{2}$$

$$\Rightarrow \quad S_b(t) = \frac{1}{2} m_p(t) \quad \text{at point b.}$$

$$S_c(t) = \frac{1}{2} m_s(t) \cdot \cos(\omega_{5.5} t)$$

$$S_d(t) = \frac{1}{2} m_s(t) \cos^2(\omega_{5.5} t)$$

$$= \frac{1}{2} m_s(t) \cdot \frac{1 + \cos(\omega_{11} t)}{2} \quad \Rightarrow S_e(t) = \frac{1}{4} m_s(t)$$

3. b. iv. If only 50 to 56 MHz frequency band is available, we should use the upper side band signal e.g., by filtering away the lower side band signal in 44 to 50 MHz before transmission.

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